Predictive Target Pursuit for Autonomous UAVs using RF-DETR with Depth-Aware State Estimation and Physics-Informed Trajectory Prediction

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DS112: Machine Learning from Challenging Data

Motivation

Autonomous UAV pursuit of airborne targets faces two compounding challenges:

- **Detection degrades** under adverse visual conditions (noise, low light, motion blur, occlusion)
- **Reactive controllers fail** when detections disappear — no input means the pursuer drifts off target

AIRHOUND addresses both within a single onboard system: a **robust transformer detector** (RF-DETR) paired with **physics-informed prediction** (PINN) that maintains tracking through multi-second dropouts.

Key question: Can architecture choice improve detection robustness under degraded conditions, and can learned prediction bridge the remaining dropout gaps?

System Architecture

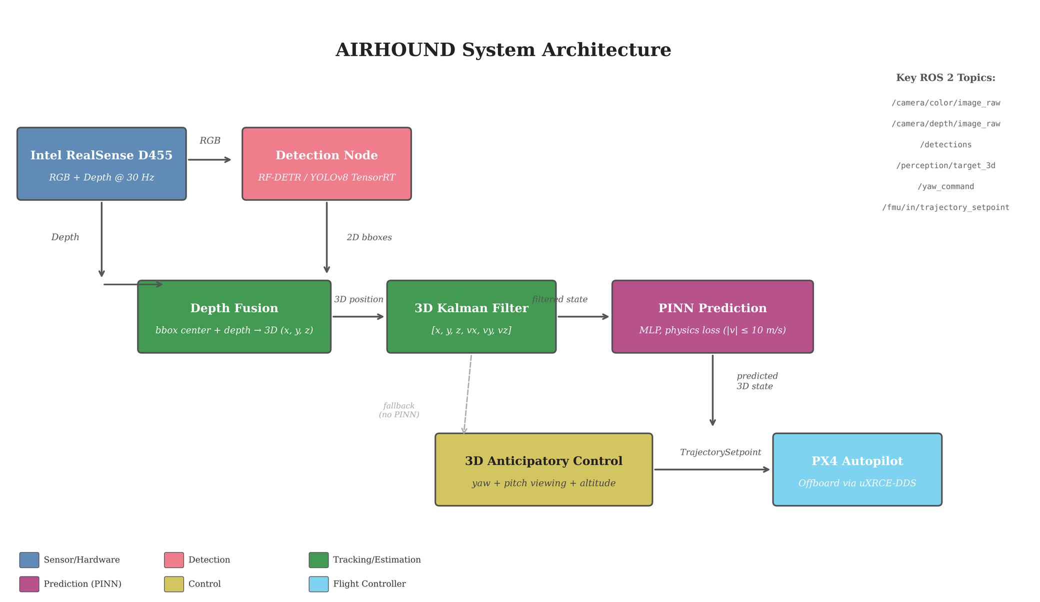

Fig. 1: AIRHOUND perception-to-control pipeline on the Jetson Orin. All nodes communicate via ROS 2 topics over Micro XRCE-DDS to the PX4 flight controller.

Hardware platform:

- NVIDIA Jetson Orin Nano (16 GB) — all inference onboard
- Intel RealSense D455 stereo camera
- Auterion PX4 FMUv6X flight controller (HIL mode)
- ROS 2 Humble with runtime YAML configuration

Physics-Informed Trajectory Prediction

During detection dropout, the PINN predicts target trajectory with a velocity-bounded physics loss:

$$\mathcal{L} = \mathcal{L}_{\text{data}} + \lambda \cdot \max\big(0,\ \|\mathbf{v}\| - v_{\max}\big)^2$$

- Prevents physically implausible extrapolation ($v_{\max} = 10$ m/s)
- TensorRT FP16 engine: **0.28 ms** inference on Jetson GPU
- Maintains tracking through dropouts up to **16.3 s** in HITL testing
- 77.5% of tracking time in PINN mode during synthetic dropout test

Degradation Robustness: RF-DETR vs. YOLOv8n

Both detectors evaluated under 4 degradation types × 5 severity levels on 1,983 validation images (Roboflow Drones dataset).

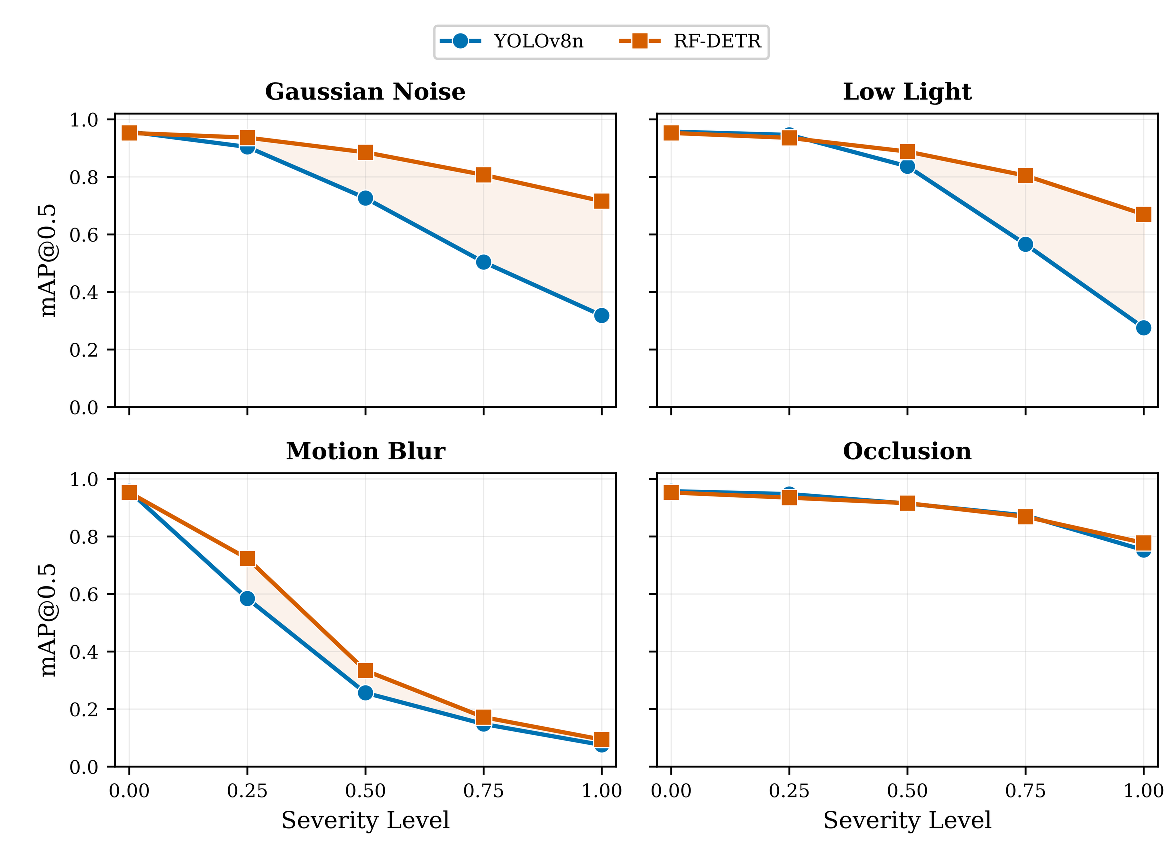

Fig. 2: Detection accuracy (mAP@0.5) under increasing degradation severity. RF-DETR (transformer) maintains substantially higher accuracy than YOLOv8n (CNN) under Gaussian noise and low-light conditions.

At maximum noise severity: RF-DETR achieves **0.716** mAP@0.5 vs. YOLOv8n at **0.318** — a **+39.8 pp** advantage

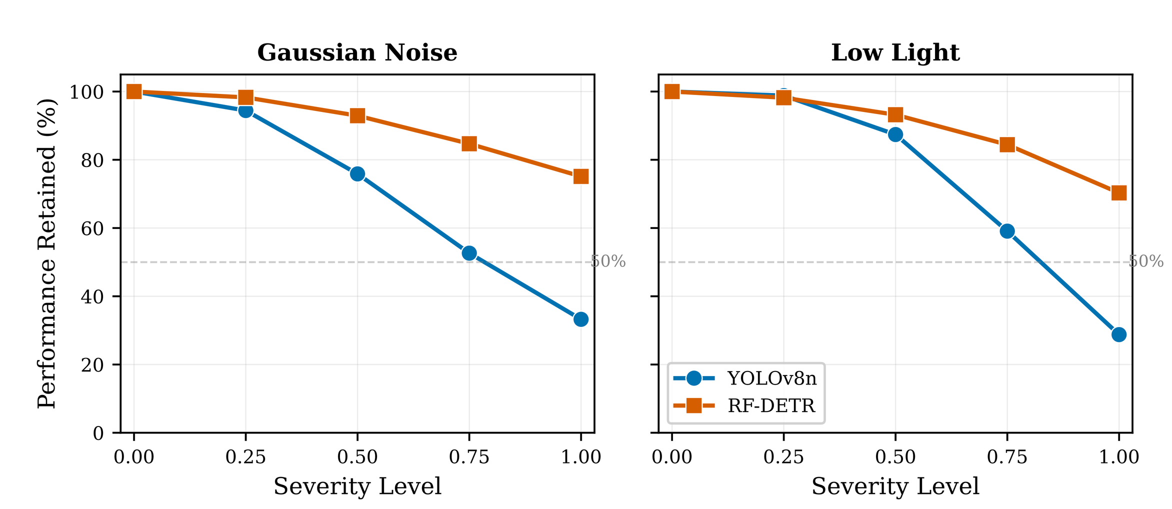

Fig. 3: Percentage of clean-baseline mAP retained under increasing severity for Gaussian noise and low light — the conditions with the largest architectural gap.

Why transformers win: Self-attention aggregates global context — uncorrupted image regions still contribute to detection even when local patches are degraded. CNN local receptive fields lose access to clean features proportionally to corruption.

References

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[4] L. Meier, D. Honegger, and M. Pollefeys, “PX4: A node-based multithreaded open source robotics framework,” *Proc. IEEE ICRA*, 2015.
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Hardware-in-the-Loop Validation

End-to-end pipeline tested with PX4 FMUv6X in HIL mode. Six configurations spanning synthetic and camera inputs.

Config	Det. Rate (%)	Latency (ms)	P95 (ms)	Dropouts
Synthetic (mock detector, simulated dropouts)				
Sim A	22.9	—	—	4
Sim B	21.9	—	—	10
Sim C†	21.0	—	—	16
Camera (RF-DETR TensorRT, RealSense D455)				
Cam 640 (bridge)	100.0	234.5	598.1	1
Cam 720 (bridge)	59.5	233.7	558.2	114
Cam Direct	**100.0**	**58.8**	**61.5**	**0**

†PINN fallback enabled; 77.5% of tracking in PINN mode.

V4L2 Direct Capture Optimization

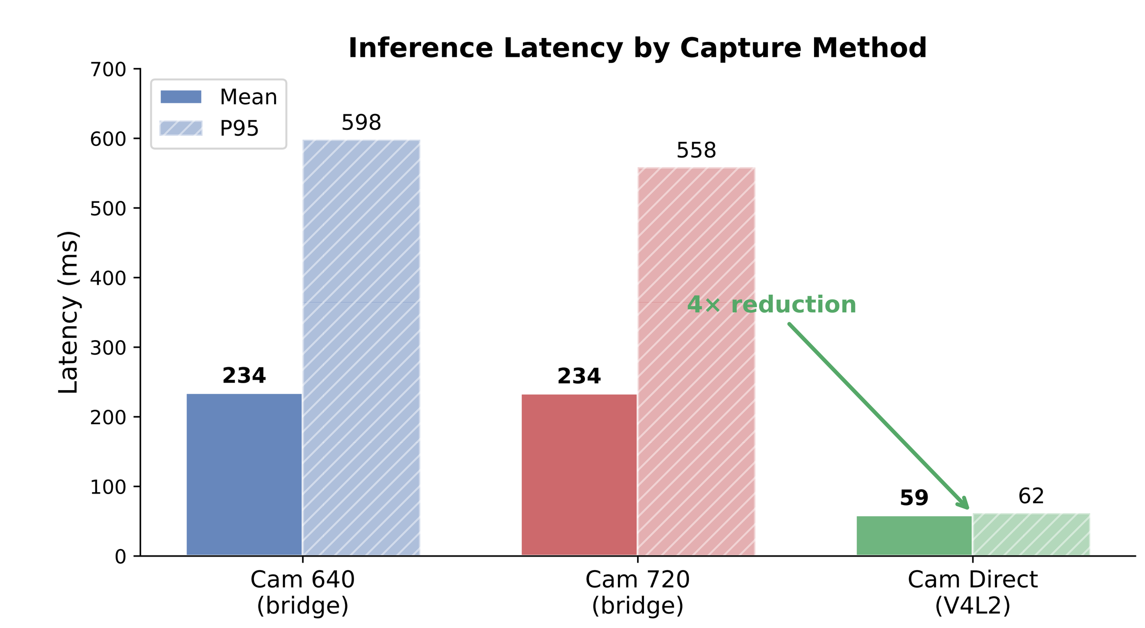

Fig. 4: Mean and P95 inference latency by capture method. V4L2 direct capture eliminates the DDS serialization bottleneck, reducing latency 4×.

ROS 2 DDS cannot serialize 2.7 MB BGR8 frames fast enough — bridge caps at ∼1.5 FPS in Python. **Integrating V4L2 capture directly into the detector node** bypasses DDS image transport:

- **4×** latency reduction (234 ms → 58.8 ms mean)
- **10×** lower latency variance (P95: 598 ms → 61.5 ms)
- **100%** detection rate at 1280×720 (was 59.5% via bridge)
- **Zero** dropouts over 108-second test

Conclusions

1. **RF-DETR** retains 75% clean mAP at max noise severity vs. 33% for YOLOv8n (+39.8 pp advantage)
2. **PINN prediction** maintains tracking through 16.3 s dropouts with velocity-bounded extrapolation
3. **Direct V4L2 capture** achieves 58.8 ms inference at 15 FPS — real-time on edge hardware
4. **Modular ROS 2 pipeline** enables runtime component selection via single YAML config

Future work: Free-flight validation, multi-axis control, PINN retraining on real trajectories.

Acknowledgments: Conducted at EPPL, Embry-Riddle Aeronautical University, under Dr. Sergey Drakunov. Supported by SPIE Student Conference Support grant, and ERAU UGR Spark grant.